

# Assignment Lab 6.2 — README

## Instructions

### Course 02 – Introduction to Machine Learning for Cybersecurity

#### Module 06: Machine Learning for Cybersecurity

### Assignment Lab 6.2

## Analyzing a Cybersecurity or Fraud Detection Scenario Using Machine Learning

### Assignment Goal

In this assignment, students will apply machine learning concepts to analyze a cybersecurity or fraud detection scenario using a real-world dataset.

Students will:

- Explore cybersecurity or fraud-related data
- Prepare data for machine learning analysis
- Train and evaluate machine learning models
- Detect suspicious or anomalous behavior
- Compare model performance
- Interpret findings in a cybersecurity context

This assignment focuses on understanding how machine learning supports cybersecurity monitoring, fraud detection, and threat analysis.

### Dataset

Use the provided dataset:

 [creditcard.csv](#)

## Dataset Description

- `Class = 0` → Normal transaction
- `Class = 1` → Fraudulent transaction

This dataset is commonly used for:

- Fraud detection
- Anomaly detection
- Suspicious behavior analysis
- Cybersecurity-focused machine learning workflows

# ✂ Assignment Tasks

## Part 1 — Import Libraries

Import the required Python and machine learning libraries.

### Suggested Libraries:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

## Part 2 — Load and Explore the Dataset

Students should:

- Load the dataset into Python
- Review dataset structure
- Check for missing values
- Analyze class distribution
- Explore suspicious behavior patterns

### **Questions to Consider:**

- Is the dataset balanced or imbalanced?
- Why is imbalance important in fraud detection?

## **Part 3 — Prepare the Data**

Students should:

- Define features ( $x$ ) and target ( $y$ )
- Split the dataset into training and testing sets
- Scale or preprocess data if needed

## **Part 4 — Train Machine Learning Models**

Students should train at least two machine learning models.

### **Suggested Models:**

- Logistic Regression
- Random Forest
- Decision Tree
- Isolation Forest (optional)

## **Part 5 — Detect Suspicious Behavior**

Students should:

- Generate predictions
- Identify suspicious or fraudulent transactions
- Compare predicted and actual outcomes

### **Questions to Consider:**

- Which model detected suspicious behavior more effectively?
- Were there many false positives or false negatives?

## Part 6 — Evaluate Model Performance

Students should calculate:

- Accuracy
- Precision
- Recall
- F1 Score

Students should also:

- Generate classification reports
- Create confusion matrices
- Compare model performance

## Part 7 — Visualize Findings

Students should create visualizations such as:

- Confusion matrix heatmaps
- Class distribution charts
- Feature importance graphs
- Fraud vs. normal transaction comparisons

## Part 8 — Interpretation and Reflection

Students should interpret:

- Suspicious activity patterns
- Model strengths and weaknesses
- Cybersecurity implications of findings
- AI limitations in cybersecurity environments



### Reflection Questions

Answer the following questions in complete sentences:

1. Which machine learning model performed best overall?

2. Why is accuracy alone not enough for fraud detection?
3. What do precision and recall tell us about suspicious behavior detection?
4. What did the confusion matrices reveal about model performance?
5. Which features appeared most important for prediction?
6. How can machine learning improve cybersecurity monitoring?
7. What are the risks of false positives and false negatives in cybersecurity systems?
8. What are some limitations of AI-driven cybersecurity systems?
9. How could this project be improved in future work?



## Submission Requirements

Submit the following in Canvas:

- ✓ Completed Jupyter Notebook (.ipynb)
- ✓ All notebook cells executed with outputs visible
- ✓ Evaluation metrics included
- ✓ Confusion matrices included
- ✓ Visualizations included
- ✓ Reflection questions answered



## Important Notes

- Update the dataset path if needed
- Use comments throughout the notebook to explain your workflow
- Focus on interpreting results, not only generating outputs
- Carefully analyze suspicious activity predictions and model behavior



## Tips for Success

- Compare multiple machine learning models carefully
- Pay close attention to precision and recall
- Think critically about false positives and false negatives

- Use visualizations to support your interpretation
- Connect findings to cybersecurity monitoring and threat detection concepts



## Cybersecurity Connection

Machine learning is increasingly used in:

- Fraud detection systems
- Intrusion detection systems (IDS)
- Threat intelligence platforms
- Security Operations Centers (SOCs)
- User behavior analytics systems

This assignment demonstrates how machine learning can help cybersecurity analysts identify suspicious activity and support data-driven security decisions.



## Final Reminder

The goal of this assignment is not only to train machine learning models, but also to understand:

- How suspicious behavior is detected through data
- How machine learning models are evaluated
- How AI supports cybersecurity operations
- How machine learning systems assist cybersecurity analysts in threat detection and monitoring